

A complex network diagram with nodes and edges, overlaid with logos and text. The nodes are small boxes containing binary strings (e.g., 0123456789, 01234567, 0123456, 012345, 01234, 0123, 012, 01, 0). The edges are colored lines (purple, green, red, blue) connecting the nodes. The diagram is centered around the text "A data structure for monomial ideals with applications to signature Gröbner basis".

*Inria*

université  
PARIS-SACLAY



# A data structure for monomial ideals with applications to signature Gröbner basis

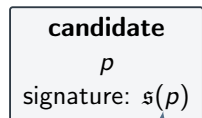
Pierre Lairez, Rafael Mohr & Théo Ternier  
MATHEXP, Inria Saclay, France

ISSAC 2026

Oldenburg, Germany – July 14, 2026

## Motivation: processing of signatures

Inside the main loop in sGB algorithm (Faugère, 2002)

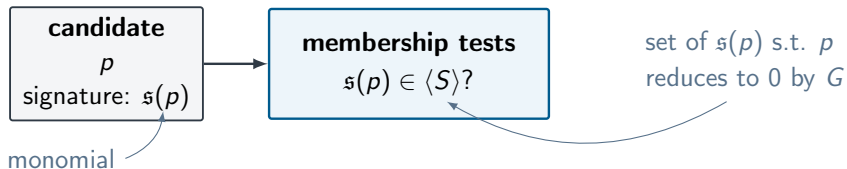


monomial



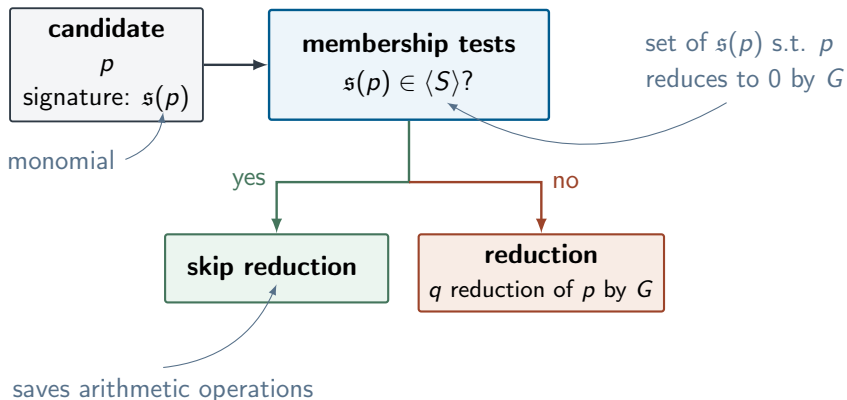
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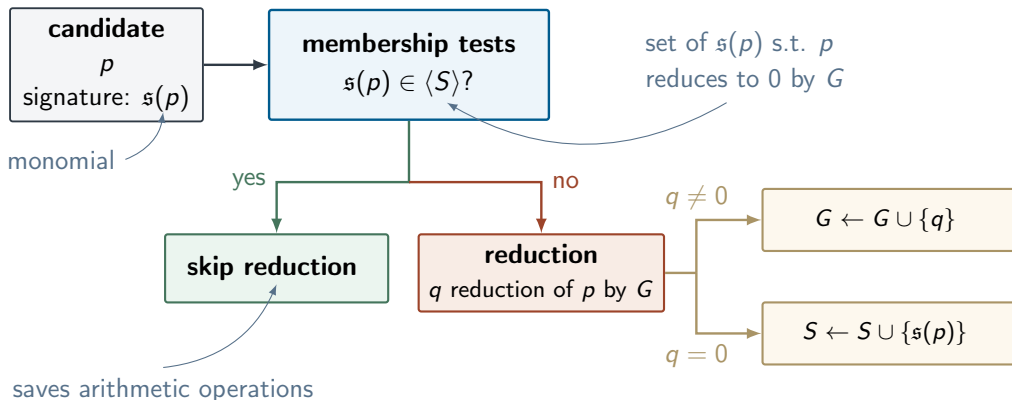
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Inside the main loop in sGB algorithm (Faugère, 2002)

candida

$p$

signature:

monomial

**eco<sup>14</sup>**: 14 polynomials in 14 variables

All reductions to 0 avoided but...

**117,861,802**

membership tests in monomial ideals

(*AlgebraicSolving.jl*)

$G \cup \{q\}$

$\cup \{s(p)\}$

saves arithmetic operations

# Membership test

**Goal:** Efficiently test monomial membership in a monomial ideal.

**Example:**  $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle$        $m = x^2y^2z^2 \in I?$

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Naive algorithm

If  $n$  number of variables and  $N$  number of generators:

$$O(nN)$$

Our data structure      Independent of  $N$

If  $d$  is the largest generator degree:

$$O(n \log d)$$

Practical impact

eco<sup>14</sup> runs 3× faster

- **Divmasks:** Prune some divisibility checks.

For each monomial  $a$  store a 64-bit integer  $\text{div}(a)$  such that for any monomial  $b$ ,

$$a \mid b \implies (\text{div}(a) \& \sim \text{div}(b)) == 0$$

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Dubé, 1993

Construction:  $O(N \log(N)^{n-1})$

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- **BDD-engine** for computations on monomial ideals.

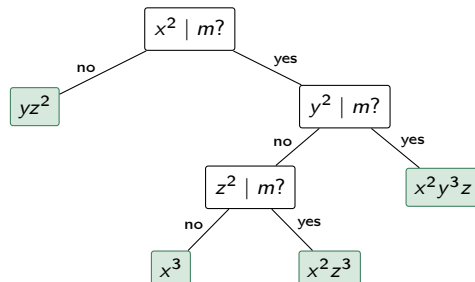
Moreno-Resa, Sáenz de Cabezón, ISSAC 2026

- Kd-trees

Roune, Stillman, ISSAC 2012

kd-tree of  $I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle$

$m = x^2y^2z^2?$



Membership:  $O(nN)$

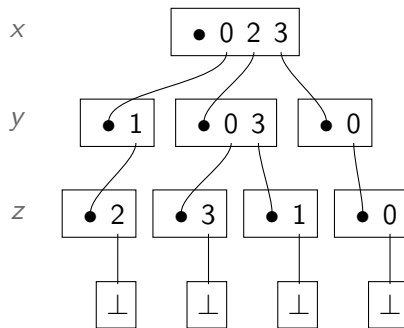
# State of the art

- Prefix tree

Milowski, 2004

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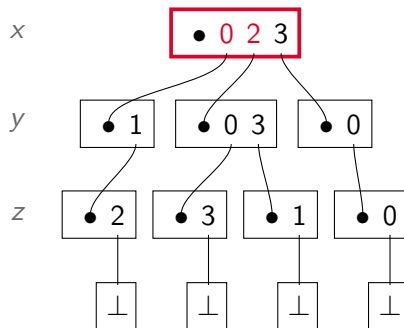
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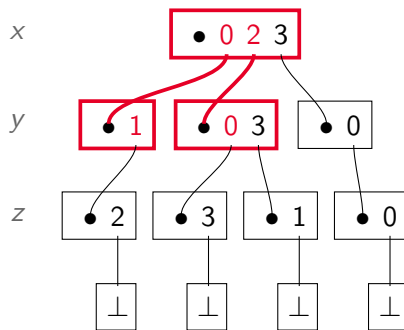
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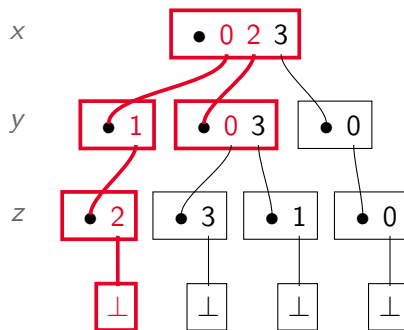
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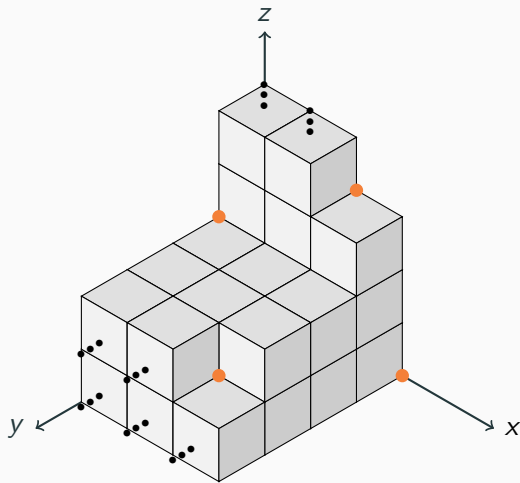


Construction:  $O(nN)$

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Geometric representation: staircase of  $I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle$

$$I/k = \langle m \in \text{Mon}(y, z) \mid x^k m \in I \rangle$$

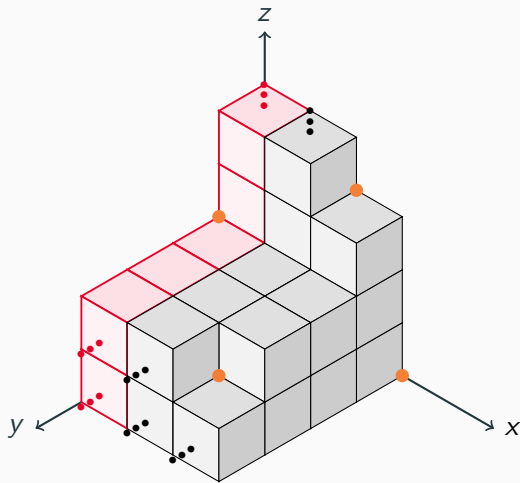




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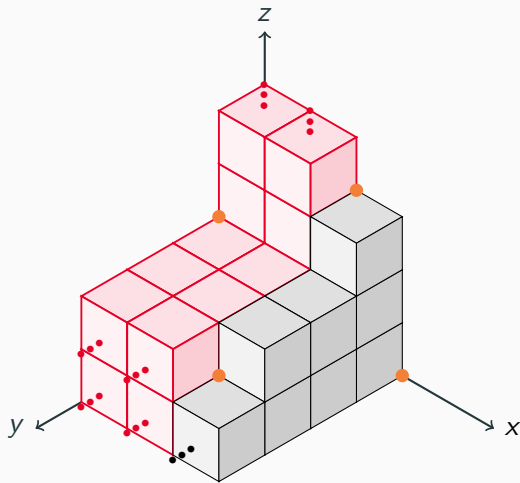
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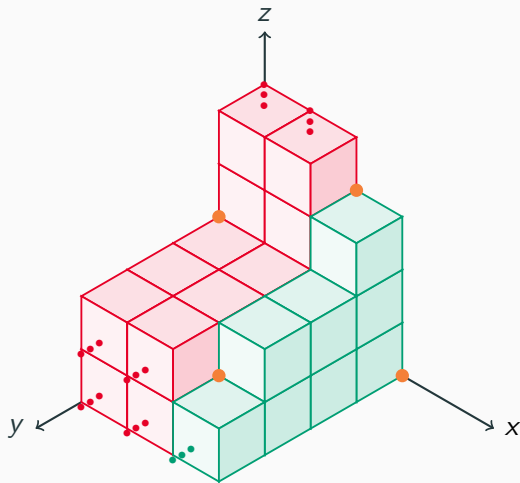
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In

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In

$$I/2 = \langle yz^2, z^3, y^3z \rangle$$



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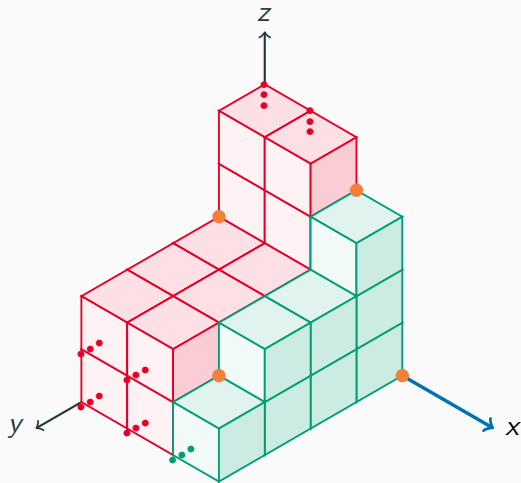
$$I/1 = \langle yz^2 \rangle$$

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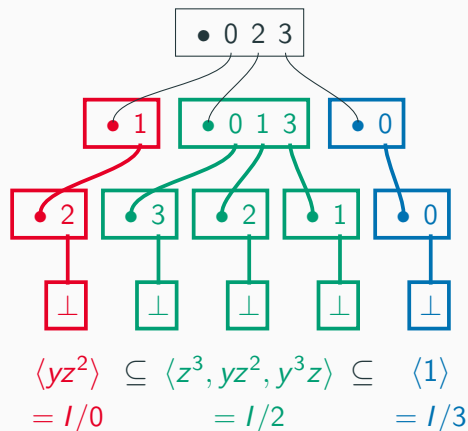
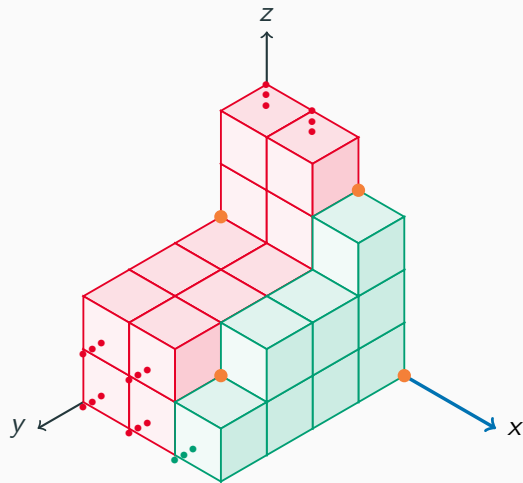
$$I/2 = \langle yz^2, z^3, y^3z \rangle$$

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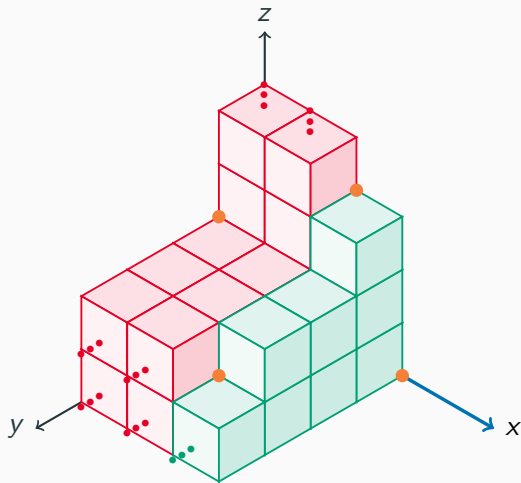
$$I/3 = \langle yz^2, z^3, y^3z, 1 \rangle$$



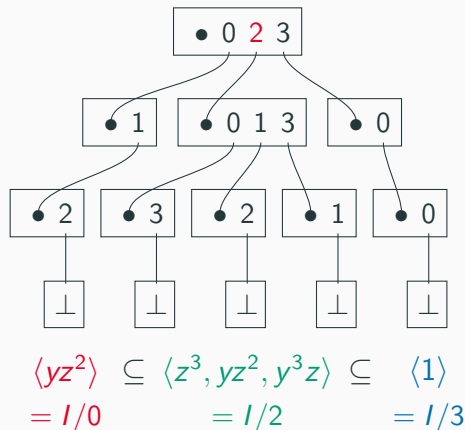
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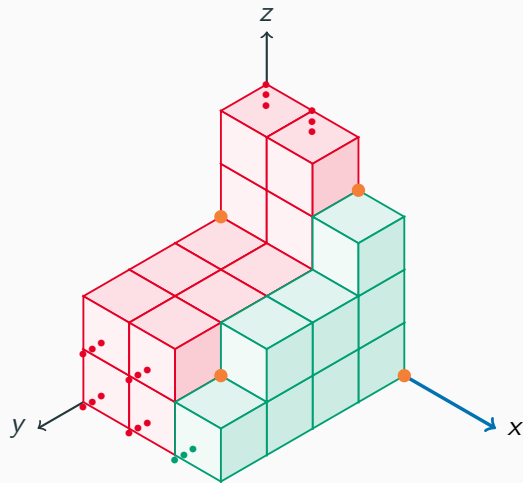
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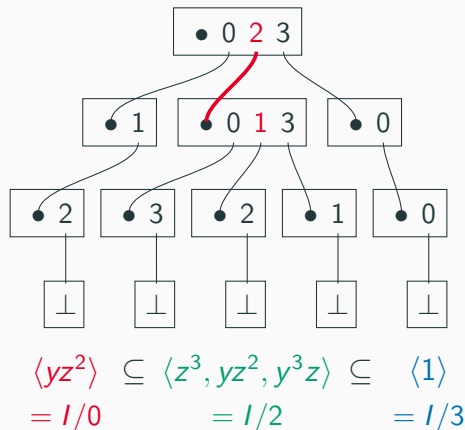
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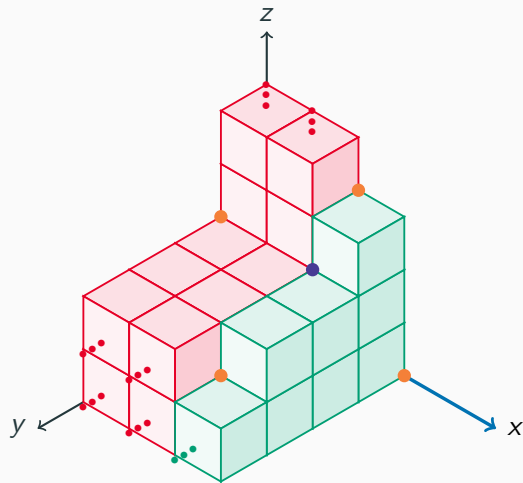
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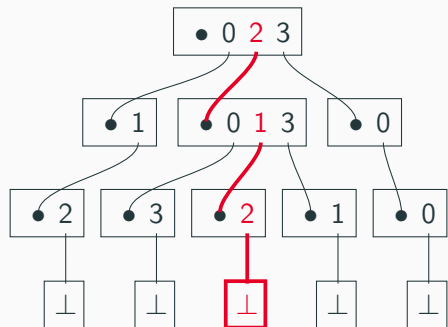
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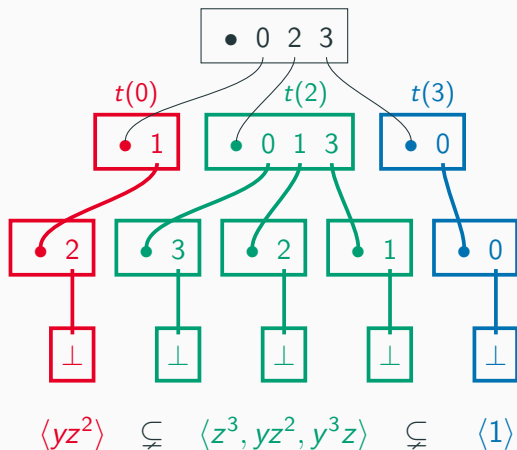
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$$\begin{aligned} \langle yz^2 \rangle &\subseteq \langle z^3, yz^2, y^3z \rangle \subseteq \langle 1 \rangle \\ &= I/0 &= I/2 &= I/3 \end{aligned}$$



# Ideal $n$ -trees

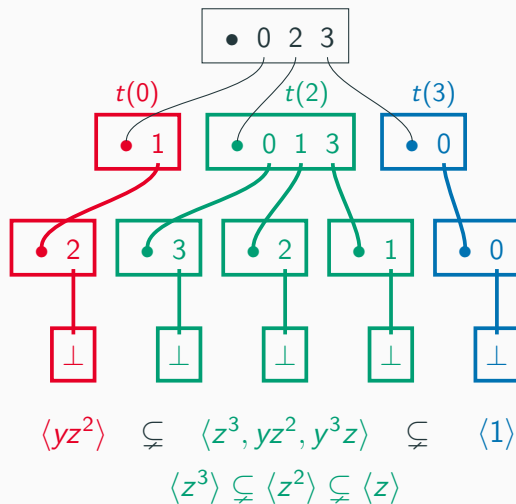


## Ideal $n$ -trees

A  $n$ -tree  $t$  is ideal if  $n = 0$  or  $n > 0$  and

- $\forall e, f \in \text{dom}(t),$   
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- $\forall e \in \text{dom}(t),$   
 $t(e)$  is ideal

## Insertion to build an ideal $n$ -tree

In sGB computations: monomials are inserted one by one

Let  $m = x_1^{m_1} \dots x_n^{m_n}$

Ideal  $n$ -tree of  $\langle m \rangle$ :



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Let  $m = x_1^{m_1} \dots x_n^{m_n}$

$I$  a monomial ideal,  $t$  its ideal  $n$ -tree. We define:

$\text{ins}(t, m)$

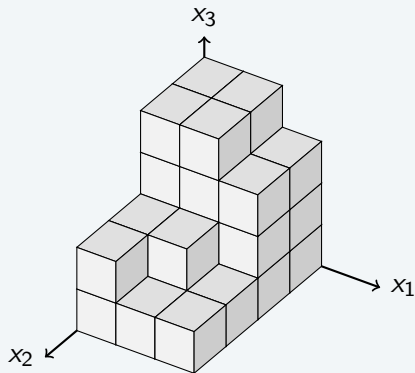
the ideal  $n$ -tree of  $I + \langle m \rangle$

Ideal  $n$ -tree of  $\langle m \rangle$ :



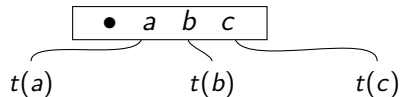
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## Geometric intuition



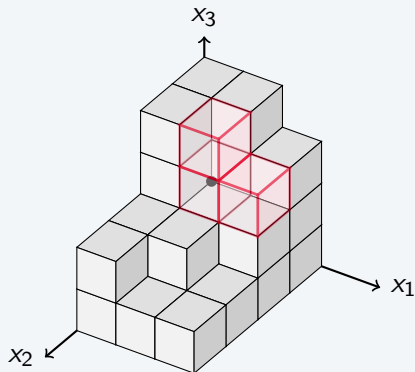
before insertion

## Tree before insertion



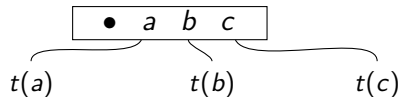
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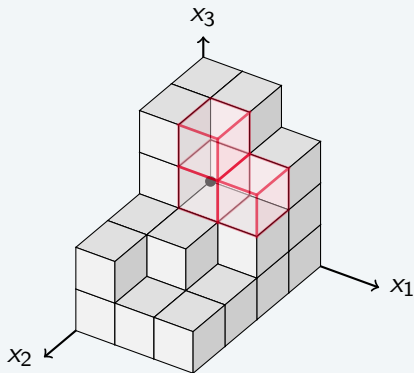
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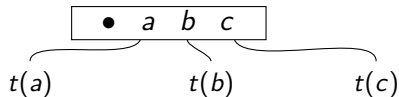
update slices  $e \geq m_1$

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$$e < m_1 : (l + \langle m \rangle) / e = l / e$$

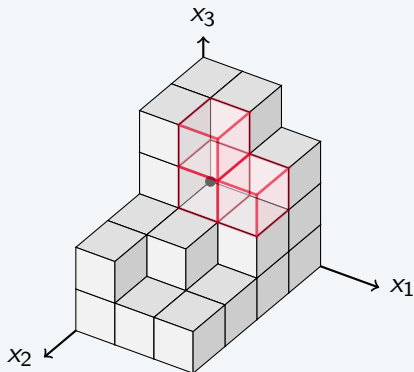
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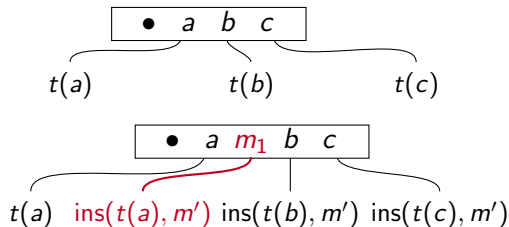
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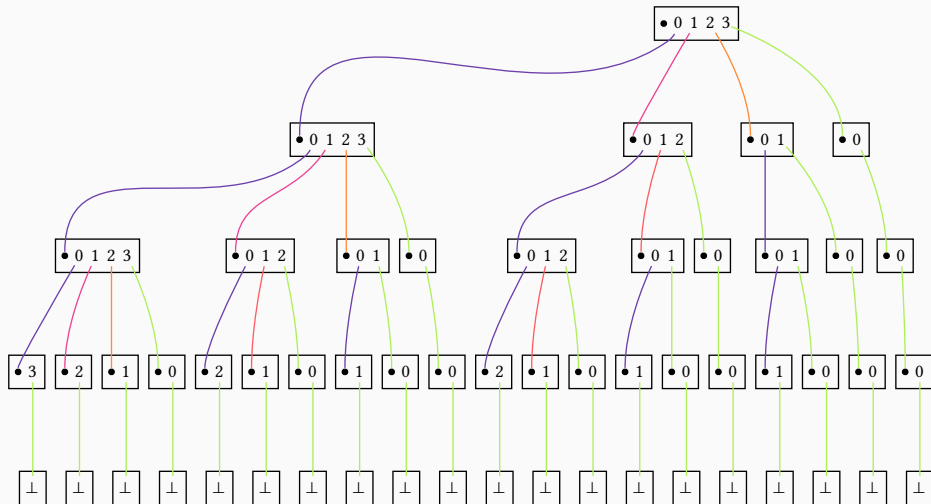
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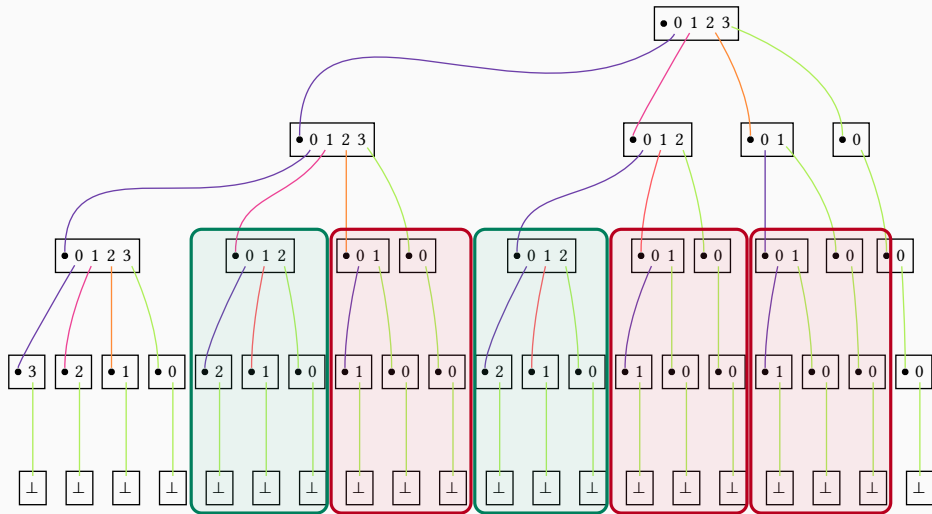




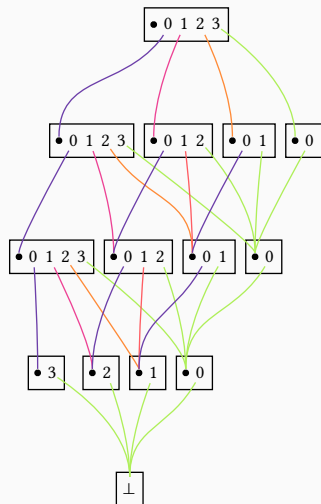
# Problem of width for ideal $n$ -trees



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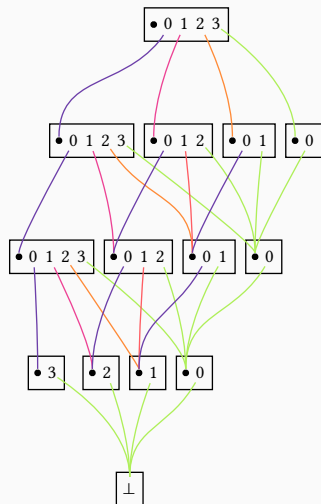
# Solved problem of width with maximal sharing



Maximal sharing

**Invariant:** *structurally equal iff physically equal.*

## Solved problem of width with maximal sharing

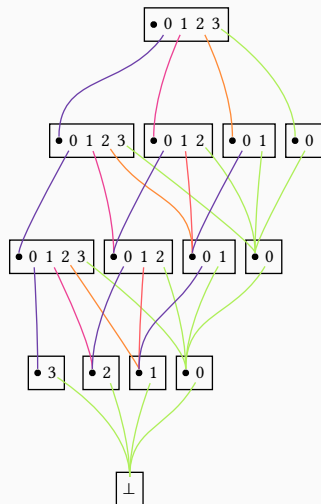


### Maximal sharing

**Invariant:** *structurally equal* iff *physically equal*.

- Identical subtrees are stored only once.

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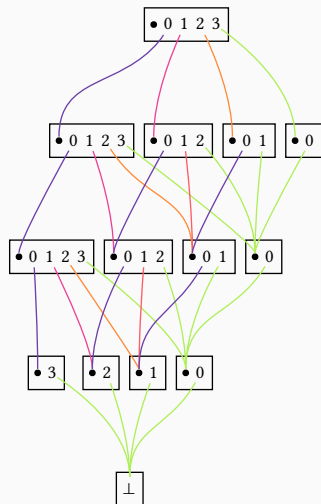


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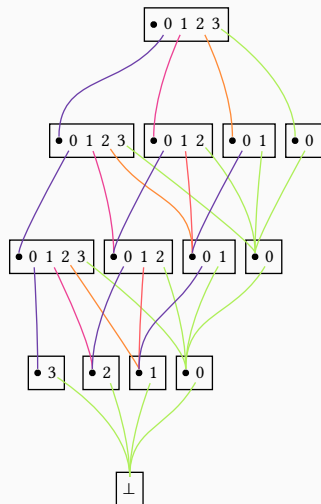


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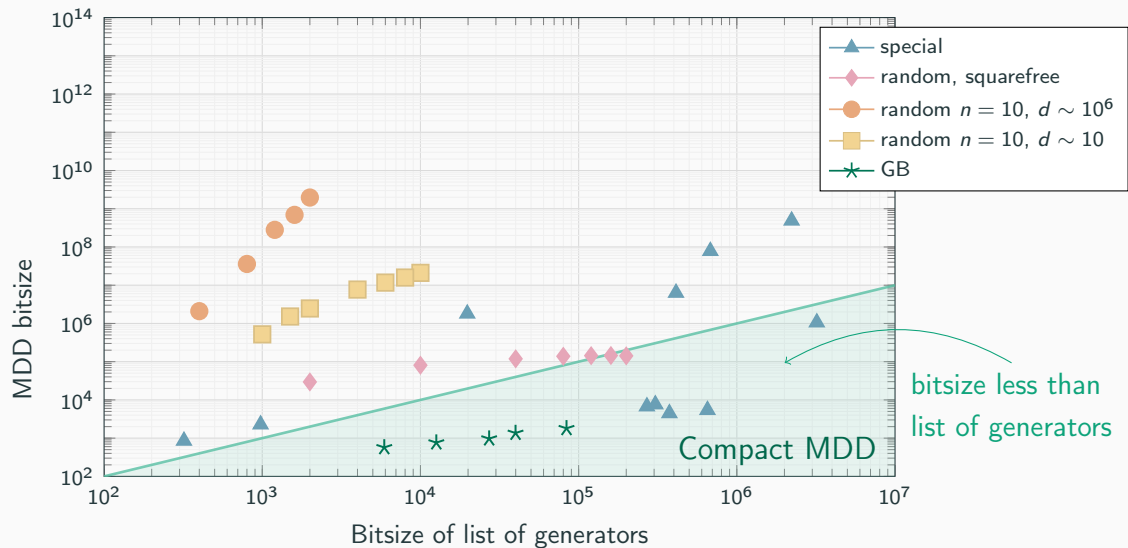
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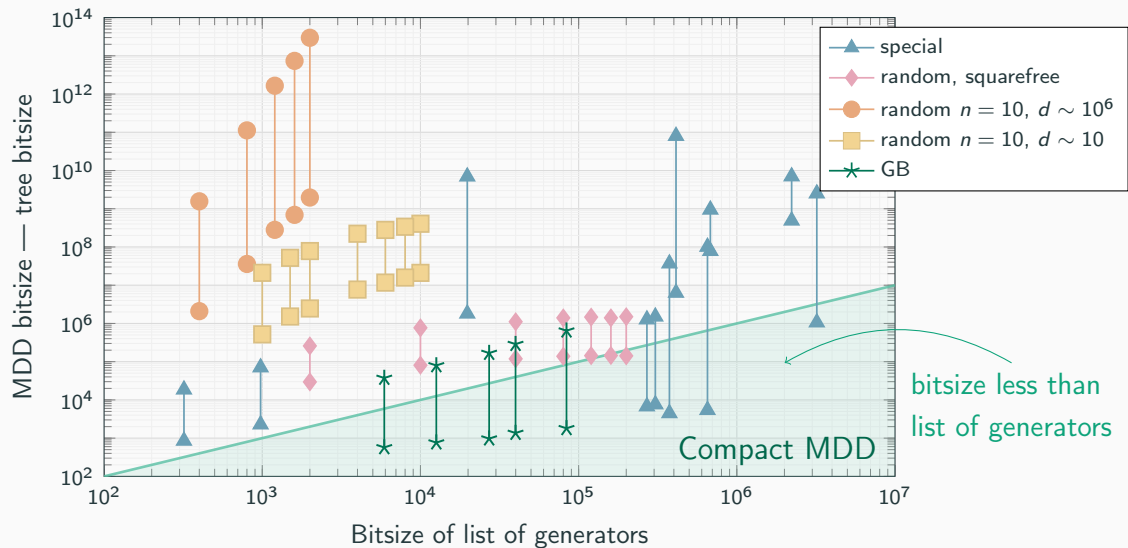
*Result: Manipulation of ideal  $n$ -trees much faster and with less memory.*

# Experimental data on the bitsize of MDDs





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# Initial ideal of $\text{eco}^{14}$ (from Gröbner basis computations)



Bitsize of list of generators: 39928

Bitsize of MDD: 1360

# Benchmarks on *AlgebraicSolving.jl*

| name                  | <i>AlgebraicSolving.jl</i> |        | <i>AlgebraicSolving.jl</i> with MDD |        |       |       |      |          |     |
|-----------------------|----------------------------|--------|-------------------------------------|--------|-------|-------|------|----------|-----|
|                       | time                       | % sig. | time                                | % sig. | test  | MDD   | size | nb tests | ×   |
| katsura <sup>11</sup> | 4s                         | 86%    | 2s                                  | 79%    | 0.04s | 0.06s | 81   | 388k     | 1.8 |
| katsura <sup>12</sup> | 27s                        | 83%    | 14s                                 | 70%    | 0.17s | 0.27s | 103  | 1.5M     | 2.0 |
| katsura <sup>13</sup> | 3m21                       | 82%    | 1m47                                | 69%    | 0.71s | 1.14s | 125  | 5.9M     | 1.9 |
| cyclic <sup>7</sup>   | 1s                         | 98%    | 0.8s                                | 98%    | 0.02s | 0.02s | 43   | 240k     | 1.6 |
| cyclic <sup>8</sup>   | 1m08                       | 95%    | 32s                                 | 89%    | 0.50s | 0.36s | 80   | 4.1M     | 2.1 |
| eco <sup>14</sup>     | 3h22                       | 90%    | 1h09                                | 68%    | 5.79s | 0.92s | 173  | 118M     | 2.9 |
| noon <sup>7</sup>     | 10s                        | 87%    | 3s                                  | 60%    | 0.08s | 0.02s | 71   | 985k     | 3.1 |
| noon <sup>8</sup>     | 2m34                       | 82%    | 44s                                 | 34%    | 0.63s | 0.10s | 101  | 7.4M     | 3.5 |
| noon <sup>9</sup>     | 1m18                       | 99%    | 1m23                                | 99%    | 0.58s | 0.10s | 71   | 7.2M     | 0.9 |

## *AlgebraicSolving.jl*

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# Benchmarks on *AlgebraicSolving.jl*

| name                  | <i>AlgebraicSolving.jl</i> |            | <i>AlgebraicSolving.jl</i> with MDD |            |              |              |            |             |            |
|-----------------------|----------------------------|------------|-------------------------------------|------------|--------------|--------------|------------|-------------|------------|
|                       | time                       | % sig.     | time                                | % sig.     | test         | MDD          | size       | nb tests    | ×          |
| katsura <sup>11</sup> | 4s                         | 86%        | 2s                                  | 79%        | 0.04s        | 0.06s        | 81         | 388k        | 1.8        |
| katsura <sup>12</sup> | 27s                        | 83%        | 14s                                 | 70%        | 0.17s        | 0.27s        | 103        | 1.5M        | 2.0        |
| katsura <sup>13</sup> | 3m21                       | 82%        | 1m47                                | 69%        | 0.71s        | 1.14s        | 125        | 5.9M        | 1.9        |
| cyclic <sup>7</sup>   | 1s                         | 98%        | 0.8s                                | 98%        | 0.02s        | 0.02s        | 43         | 240k        | 1.6        |
| cyclic <sup>8</sup>   | 1m08                       | 95%        | 32s                                 | 89%        | 0.50s        | 0.36s        | 80         | 4.1M        | 2.1        |
| eco <sup>14</sup>     | <b>3h22</b>                | <b>90%</b> | <b>1h09</b>                         | <b>68%</b> | <b>5.79s</b> | <b>0.92s</b> | <b>173</b> | <b>118M</b> | <b>2.9</b> |
| noon <sup>7</sup>     | 10s                        | 87%        | 3s                                  | 60%        | 0.08s        | 0.02s        | 71         | 985k        | 3.1        |
| noon <sup>8</sup>     | 2m34                       | 82%        | 44s                                 | 34%        | 0.63s        | 0.10s        | 101        | 7.4M        | 3.5        |
| noon <sup>9</sup>     | 1m18                       | 99%        | 1m23                                | 99%        | 0.58s        | 0.10s        | 71         | 7.2M        | 0.9        |

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## MDD bitsize

Initial ideals from GB computations:

- Generic coordinates:  
Borel-fixed (Galligo, 1974)

$$O(ndN)$$

- Generic systems: conjectured to be  
Weakly revlex (Moreno–Socias, 1991)

$$O(nd^2)$$

## Find reducer

Go beyond yes/no membership:

$m \in I \rightsquigarrow$  minimal generator index

Improve the symbolic part

## Applications in commutative algebra

Hilbert series